

Computer Organization Project 1 – MIPS Assembly 1

Due: 23:59, Mar. 25, 2026

In this start-up project, you are required to get familiar with both MARS (MIPS Assembler and Runtime Simulator) programming environment and basic MIPS (Microprocessor without Interlocked Pipeline Stages) instructions.

The Roman numeral system originated in ancient Rome and remained the usual way of writing numbers throughout Europe well into the Late Middle Ages. In this project, you are required to convert a decimal integer (1~3999) into the corresponding Roman numeral representation.

Symbol	I	V	X	L	C	D	M
Number	1	5	10	50	100	500	1000

The Roman Numeral Rules are described as follows:

- (1) Additive rule (typically written in descending order): If a symbol is followed by a symbol of equal or smaller value, the values are added.
Example: VIII = 5 + 1 + 1 + 1 = 8, LX = 50 + 10 = 60.
- (2) Subtractive rule (limited cases only): A smaller value placed before a larger value indicates subtraction, and only the following pairs are allowed: IV = 4, IX = 9, XL = 40, XC = 90, CD = 400, CM = 900.
- (3) Repetition rule: I, X, C, M can be repeated up to three times in a row (e.g., III, XXX, CCC, MMM). V, L, D must not be repeated.
- (4) Valid input range: This project only needs to support 1~3999.

You are required to write a program to convert a decimal integer to the corresponding Roman numeral according to the above table and rules. Please submit your source code according to the following rules:

- 1- Write down enough comments such that you would receive higher scores.
- 2- The filename is your student ID (e.g., B12345678.asm).

Example:

```
Please input the decimal integer (1~3999):  
1994  
The Roman numeral is MCMXCIV.
```

Please input the decimal integer (1~3999):
58
The Roman numeral is LVIII.

Please input the decimal integer (1~3999):
3289
The Roman numeral is MMMCCLXXXIX.

Please input the decimal integer (1~3999):
2026
The Roman numeral is MMXXVI.

Please input the decimal integer (1~3999):
0
Illegal input.

Please input the decimal integer (1~3999):
4000
Illegal input.